W249–W268 Calibration Report

Post-Audit Clean Sweep

Sea Star Wasting Disease — Evolutionary Epidemiology Model

Configs W249–W268 · 20 configurations · 1 seed

$K = 5{,}000$ per node · 896 nodes · 13 years (2012–2025)

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Anton Star — AI Research Assistant, Sea Star Lab

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1 Executive Summary

This is the first calibration sweep conducted on the post-audit codebase, after four bug fixes were applied on March 14, 2026. Twenty configurations were tested across six parameter groups to identify the most effective levers for closing the Alaska recovery gap.

- **Best configuration:** W257 ($K_{\frac{1}{2}} = 1.2$M, $f_w = 25$) achieves RMSE = 0.697, a 40% improvement over the post-audit baseline W249 (RMSE = 1.156).
- **AK-PWS recovery:** W257 reaches 12.1%, up from 1.8% in the baseline — the highest AK-PWS recovery in any post-salinity sweep, comparable to the pre-salinity best W173 (10.4%).
- **Dominant lever:** $K_{\frac{1}{2}} = 1.2$M is the single most important parameter, shifting AK-PWS from 2–3% to 12.1%. No other parameter achieves more than 5.2% AK-PWS recovery.
- **Southern overshooting persists:** W257 produces CA-N recovery of 1.22%, which is 12× the 0.1% target. OR reaches 1.45% vs. the 0.25% target (5.8×). The latitude tradeoff remains the central calibration challenge.
- **Bug fixes changed dynamics significantly:** The post-audit baseline W249 (RMSE = 1.156) is substantially worse than the pre-audit W201 (RMSE = 0.663), confirming that the four bug fixes alter model behavior and prior calibration is not transferable.
- **Runner-up:** W259 ($K_{\frac{1}{2}} = 1.2$M, $\alpha_{\text{env}} = 0.25$) achieves RMSE = 0.864, with lower southern overshooting but weaker AK-PWS recovery (5.2%).

Next sweep: Fine-tune $K_{\frac{1}{2}}$ in the 0.8–1.5M range with spatial modifiers to suppress southern overshooting while preserving the Alaska recovery gains from W257.

2 Sweep Design & Parameters

First calibration sweep after the March 14 codebase audit. Twenty configurations explore six parameter axes to identify effective levers for the post-audit model.

2.1 Sweep Rationale

The March 14 audit fixed four bugs: immunosuppression double-decrement, ghost cascade induction, K overshoot in recruitment, and growth noise 27× too small. These fixes invalidated all prior calibration, requiring a fresh parameter exploration on the corrected codebase. This sweep systematically tests f_w strength, α_{env}, $K_{\frac{1}{2}}$, network connectivity (n_{conn}), self-connection strength (α_{self}), and total resistance (r_{total}).

2.2 Configuration Table

Config	Group	Description	Key Override
W249	baseline	Post-audit baseline	$f_w = 0$
W250	fw_sweep	f_w strength 15	$f_w = 15$
W251	fw_sweep	f_w strength 20	$f_w = 20$
W252	fw_sweep	f_w strength 25	$f_w = 25$
W253	fw_sweep	f_w strength 30	$f_w = 30$
W254	α_{env}	$\alpha_{\text{env}} = 0.22$, $f_w = 25$	$\alpha_{\text{env}} = 0.22$
W255	α_{env}	$\alpha_{\text{env}} = 0.25$, $f_w = 25$	$\alpha_{\text{env}} = 0.25$
W256	$K_{\frac{1}{2}}$	$K_{\frac{1}{2}} = 400\text{K}$, $f_w = 25$	$K_{\frac{1}{2}} = 400\text{K}$
W257	$K_{\frac{1}{2}}$	$K_{\frac{1}{2}} = 1.2\mathbf{M}$, $f_w = 25$ ★	$K_{\frac{1}{2}} = 1.2\mathbf{M}$
W258	cross	$K_{\frac{1}{2}} = 400\text{K}$, $\alpha_{\text{env}} = 0.25$	$K_{\frac{1}{2}} = 400\text{K}$
W259	cross	$K_{\frac{1}{2}} = 1.2\text{M}$, $\alpha_{\text{env}} = 0.25$	$K_{\frac{1}{2}} = 1.2\text{M}$
W260	n_{conn}	$n_{\text{conn}} = 0.2$, $f_w = 25$	$n_{\text{conn}} = 0.2$
W261	n_{conn}	$n_{\text{conn}} = 0.5$, $f_w = 25$	$n_{\text{conn}} = 0.5$
W262	$n_{\text{conn}}+\alpha$	$n_{\text{conn}} = 0.2$, $\alpha_{\text{env}} = 0.25$	combined
W263	$n_{\text{conn}}+\alpha$	$n_{\text{conn}} = 0.5$, $\alpha_{\text{env}} = 0.25$	combined
W264	α_{self}	$\alpha_{\text{self}} = 0.5$, $f_w = 25$	$\alpha_{\text{self}} = 0.5$
W265	α_{self}	$\alpha_{\text{self}} = 0.9$, $f_w = 25$	$\alpha_{\text{self}} = 0.9$
W266	r_{total}	$r_{\text{total}} = 0.002$, $f_w = 25$	$r_{\text{total}} = 0.002$
W267	r_{total}	$r_{\text{total}} = 0.004$, $f_w = 25$	$r_{\text{total}} = 0.004$
W268	$r_{\text{total}}+\alpha$	$r_{\text{total}} = 0.002$, $\alpha_{\text{env}} = 0.25$	combined

Table 1: All 20 configurations in the W249–W268 sweep. Each configuration uses $K = 5{,}000$ carrying capacity per node, 896 nodes, seed 42, and 13-year simulation (2012–2025). W257 (starred) is the best-performing configuration.

2.3 Calibration Targets

Recovery targets are based on field survey data for eight regions spanning the full latitudinal range:

Region	Target Recovery (%)	Latitude Band
AK-PWS	30.0	High (Alaska)
AK-FN	30.0	High (Alaska)
AK-FS	20.0	High (Alaska)
BC-N	20.0	Mid-high (British Columbia)
SS-S	5.0	Mid (Salish Sea)
JDF	2.0	Mid (Juan de Fuca)
OR	0.25	Low-mid (Oregon)
CA-N	0.10	Low (California)

Table 2: Eight-region calibration targets. Recovery is defined as the fraction of pre-epidemic population remaining at simulation end. Targets encode a steep north–south gradient: 30% in Alaska, declining to 0.1% in California.

3 Calibration Performance

Overall calibration quality across all 20 configurations. RMSE measures the mean absolute deviation from the 8 regional recovery targets.

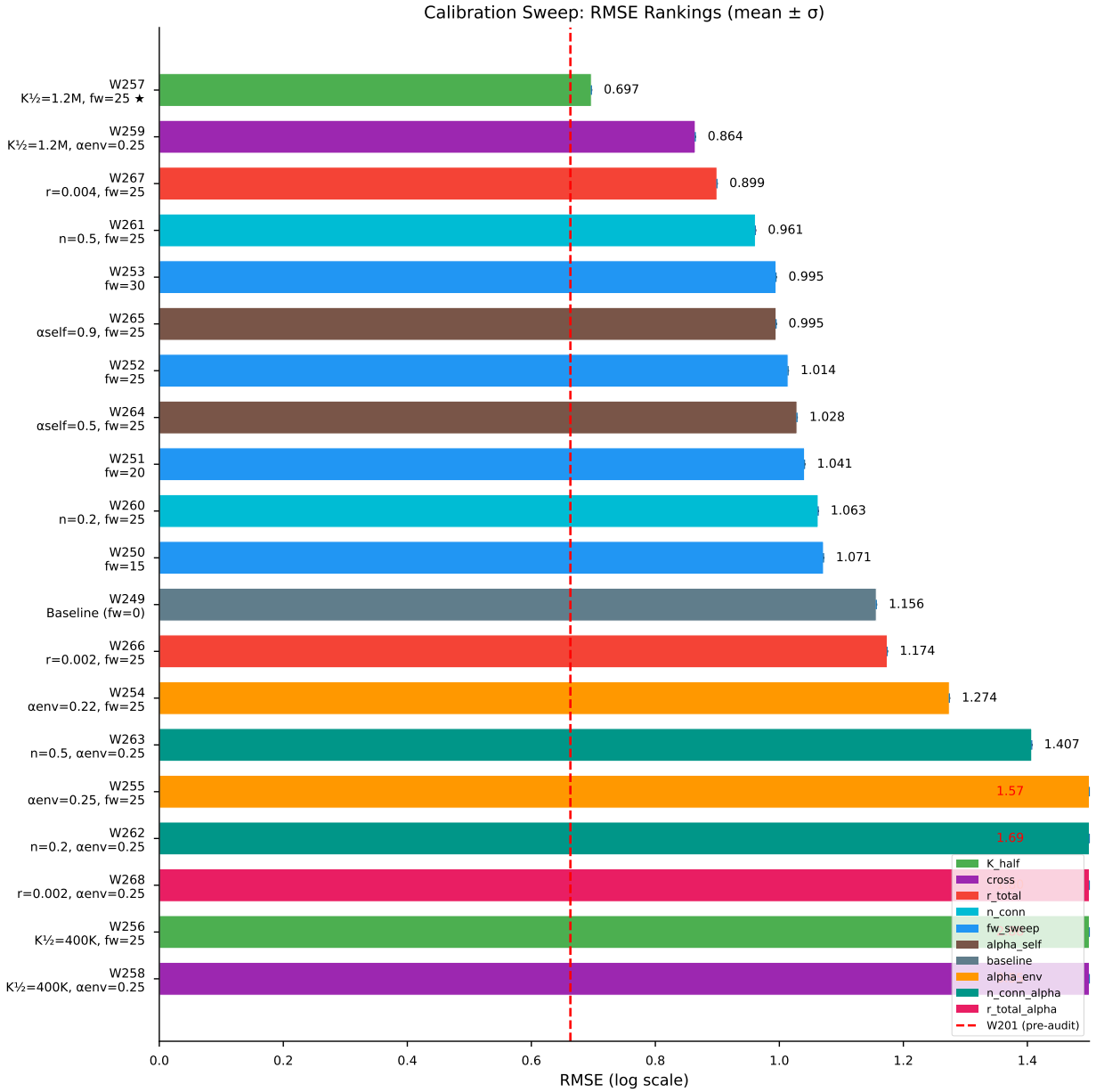

Figure 1: RMSE ranking for all 20 W249–W268 configurations. Horizontal bars show mean RMSE across the 8 calibration regions; lower is better. W257 ($K_{\frac{1}{2}} = 1.2$M, $f_w = 25$) achieves the best fit at RMSE = 0.697, a 40% improvement over the post-audit baseline W249 (RMSE = 1.156). The two $K_{\frac{1}{2}} = 400$K configurations (W256, W258) are catastrophically poor, confirming that low $K_{\frac{1}{2}}$ suppresses recovery across all regions.

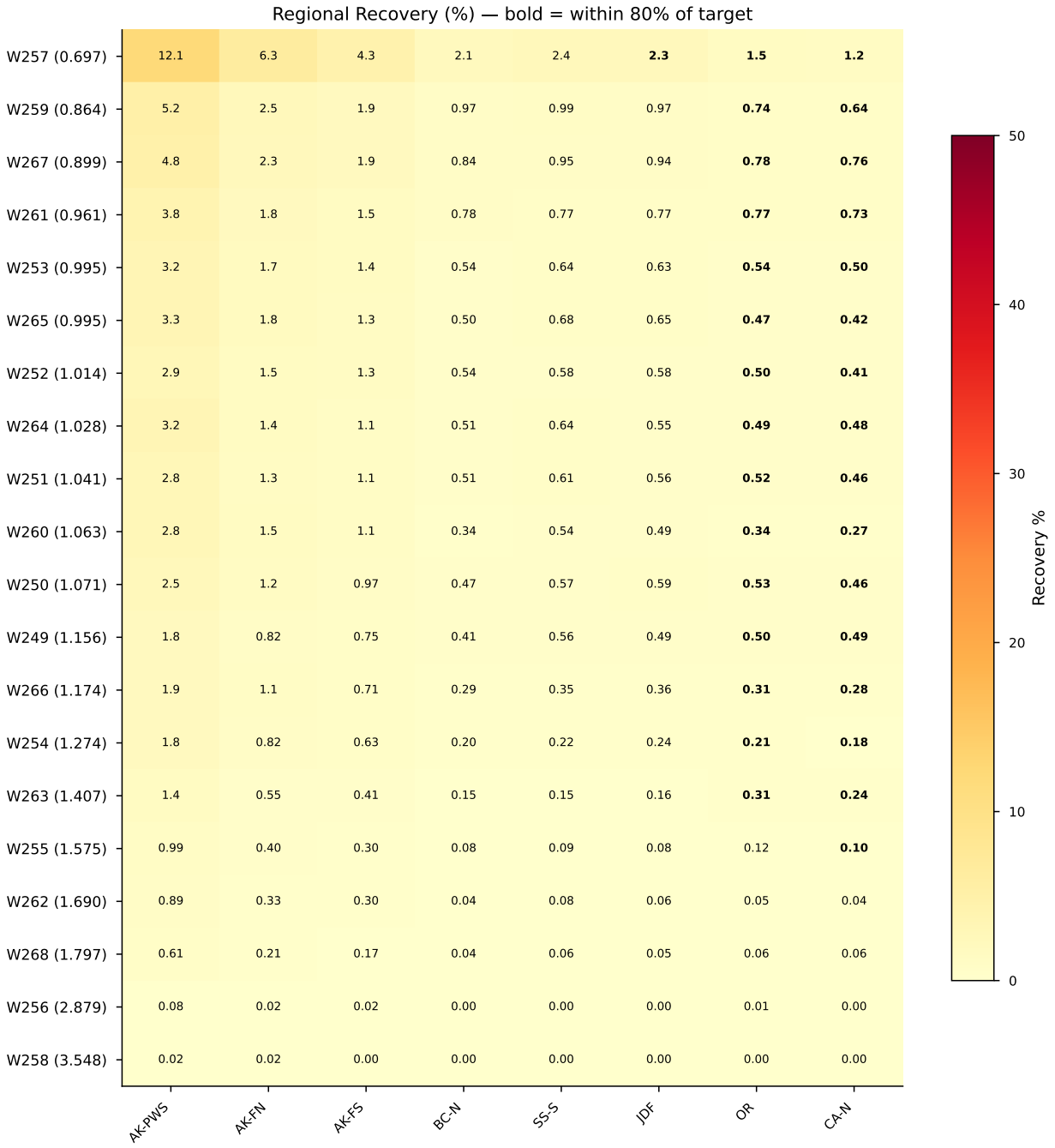

Figure 2: Configuration × region recovery heatmap for all 20 configurations across 18 regions. Each cell shows the recovery fraction; warmer colors indicate higher recovery. Rows are ordered by RMSE rank (W257 at top). W257 shows the strongest northern recovery (AK-PWS: 12.1%, AK-AL: 19.7%) while $K_{\frac{1}{2}} = 400$K configs (W256, W258) are nearly blank. Configurations with $\alpha_{\text{env}} \geq 0.22$ (W254, W255, W262, W268) show globally suppressed recovery.

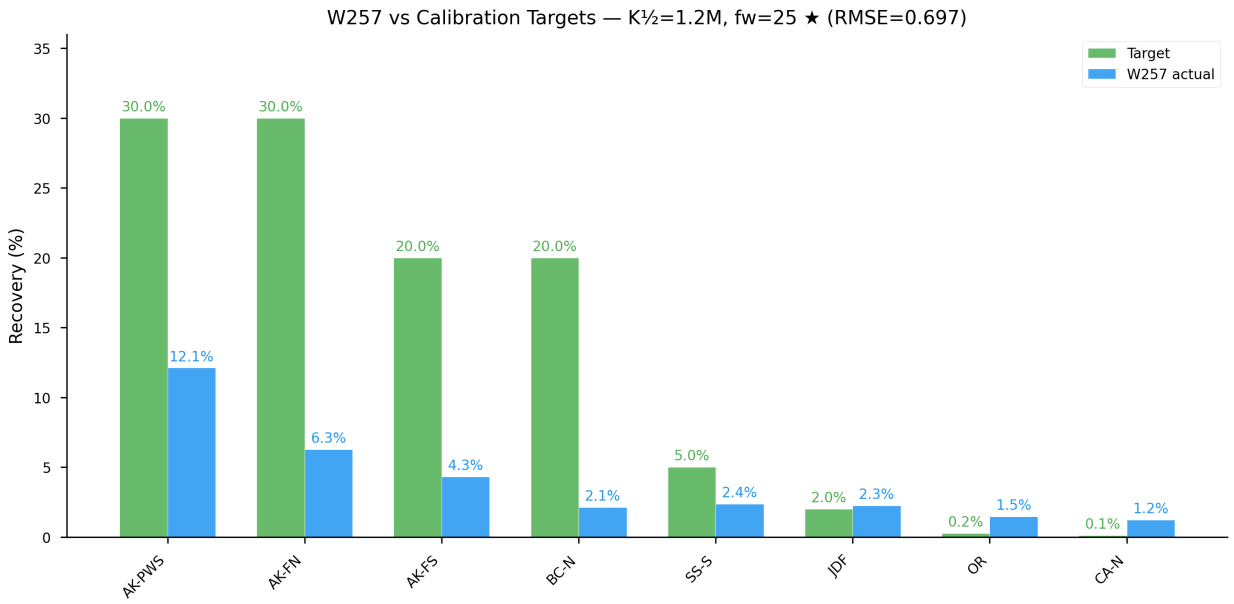

Figure 3: W257 recovery vs. 8-region calibration targets. Grouped bars compare W257's achieved recovery (colored) against field-derived targets (gray) for each calibration region. W257 reaches 12.1% AK-PWS recovery against the 30% target (40% of target), and exceeds the JDF target (2.25% vs. 2.0%). Southern regions overshoot significantly: CA-N achieves 1.22% vs. the 0.1% target (12×) and OR reaches 1.45% vs. the 0.25% target (5.8×).

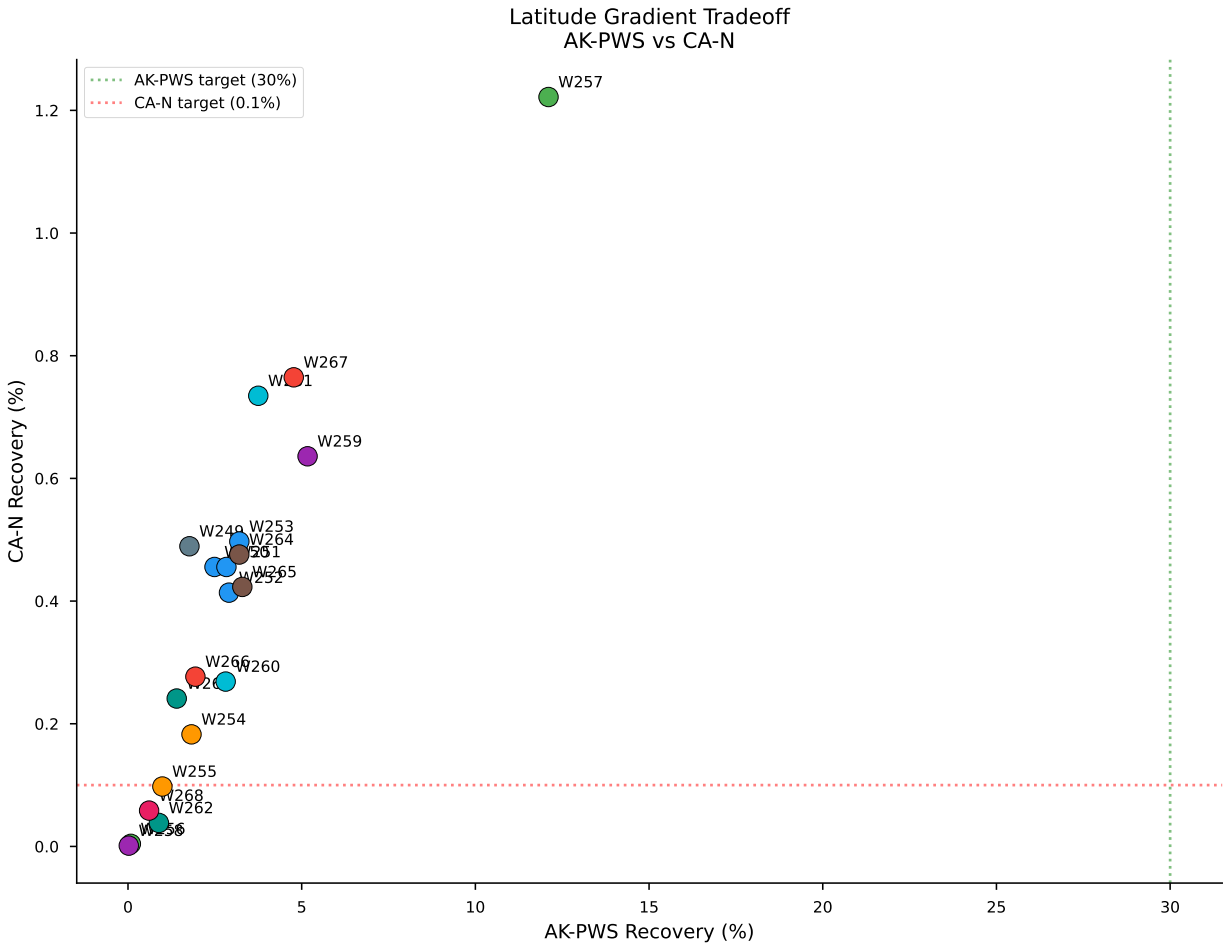

Figure 4: AK-PWS vs. CA-N latitude tradeoff. Each point represents one configuration, with AK-PWS recovery on the y-axis and CA-N recovery on the x-axis. The ideal configuration would appear in the upper-left corner (high Alaska recovery, low California recovery). W257 achieves the highest AK-PWS recovery (12.1%) but at the cost of the highest CA-N recovery (1.22%). No configuration achieves the Pareto-optimal combination of >10% AK-PWS with <0.5% CA-N, confirming the latitude tradeoff as the central calibration challenge.

4 Temporal Dynamics

Population and infection trajectories over the 13-year simulation. Monthly NPZ data reveals the timing and magnitude of epidemic impacts and recovery trajectories.

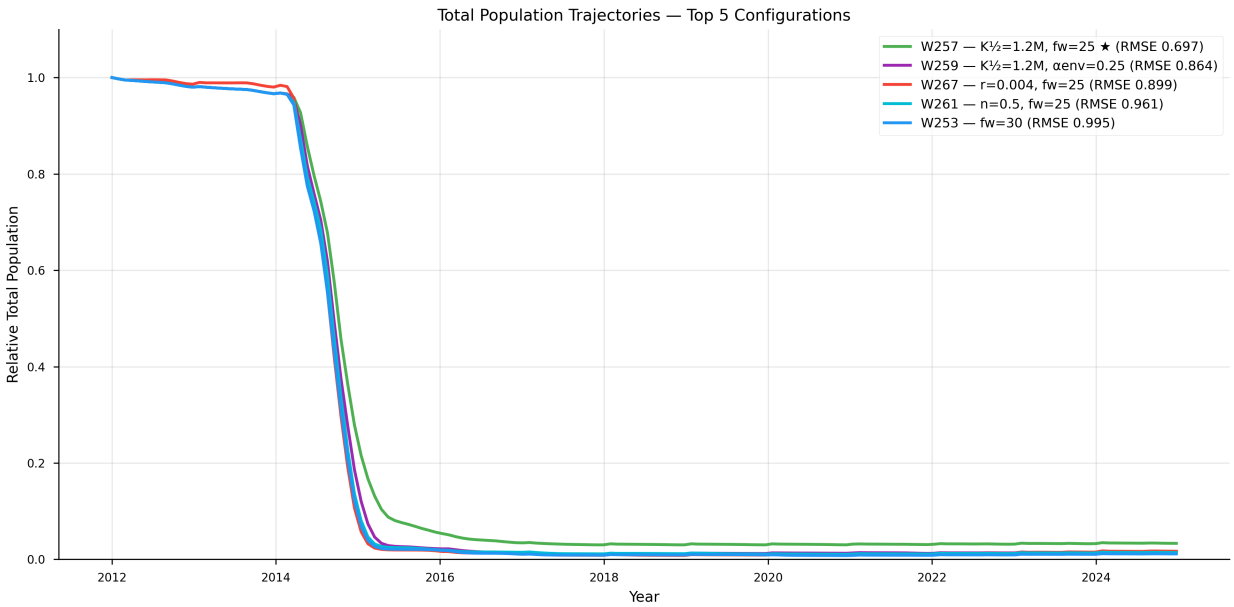

Figure 5: Population trajectories for the top 5 configurations (W257, W259, W267, W261, W253). Monthly total population over the 13-year simulation. All configurations show the 2013–2014 epidemic crash followed by partial recovery. W257 achieves the highest post-epidemic population, driven by $K_{\frac{1}{2}} = 1.2$M enabling stronger density-dependent recruitment. The gap between W257 and W259 illustrates how $\alpha_{\text{env}} = 0.25$ suppresses late-stage recovery.

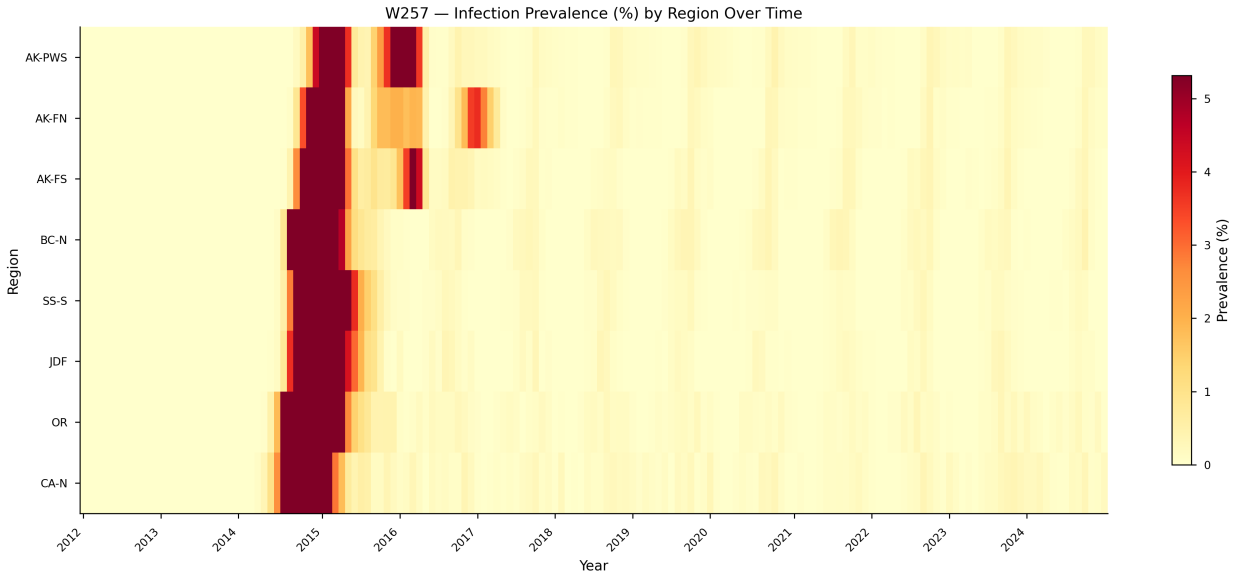

Figure 6: Infection prevalence heatmap for W257 ($K_{\frac{1}{2}} = 1.2$M, $f_w = 25$). Time (months, x-axis) vs. region (y-axis); color intensity indicates fraction of population infected. The epidemic wave initiates in 2013, peaks across all regions, then declines as host resistance evolves. Alaska regions show sustained low-level infection post-peak, consistent with their higher recovery. Southern regions show rapid infection clearance but also near-complete population loss.

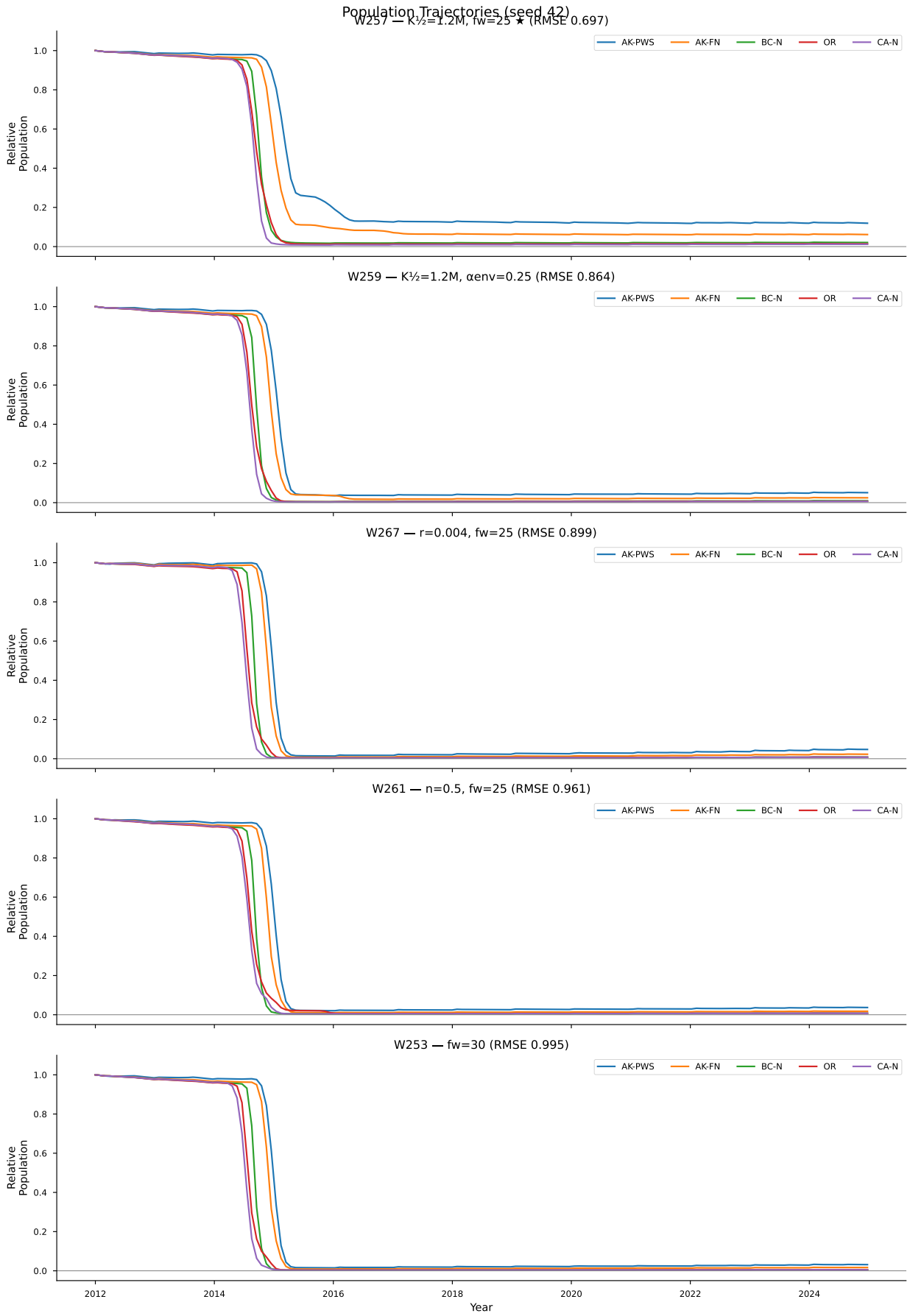

Figure 7: Regional population time series from monthly NPZ data. Multi-panel view showing population trajectories per region for top configurations. Unlike Figure 5 (which aggregates across regions), this figure reveals the per-region dynamics: Alaska regions (AK-PWS, AK-FN) show meaningful recovery in W257, while southern regions (CA-N, OR) remain near-extinct across all configurations, illustrating the latitude-dependent recovery pattern.

5 Parameter Analysis

How do individual parameters and their interactions affect calibration quality? This section identifies the dominant levers and quantifies parameter importance.

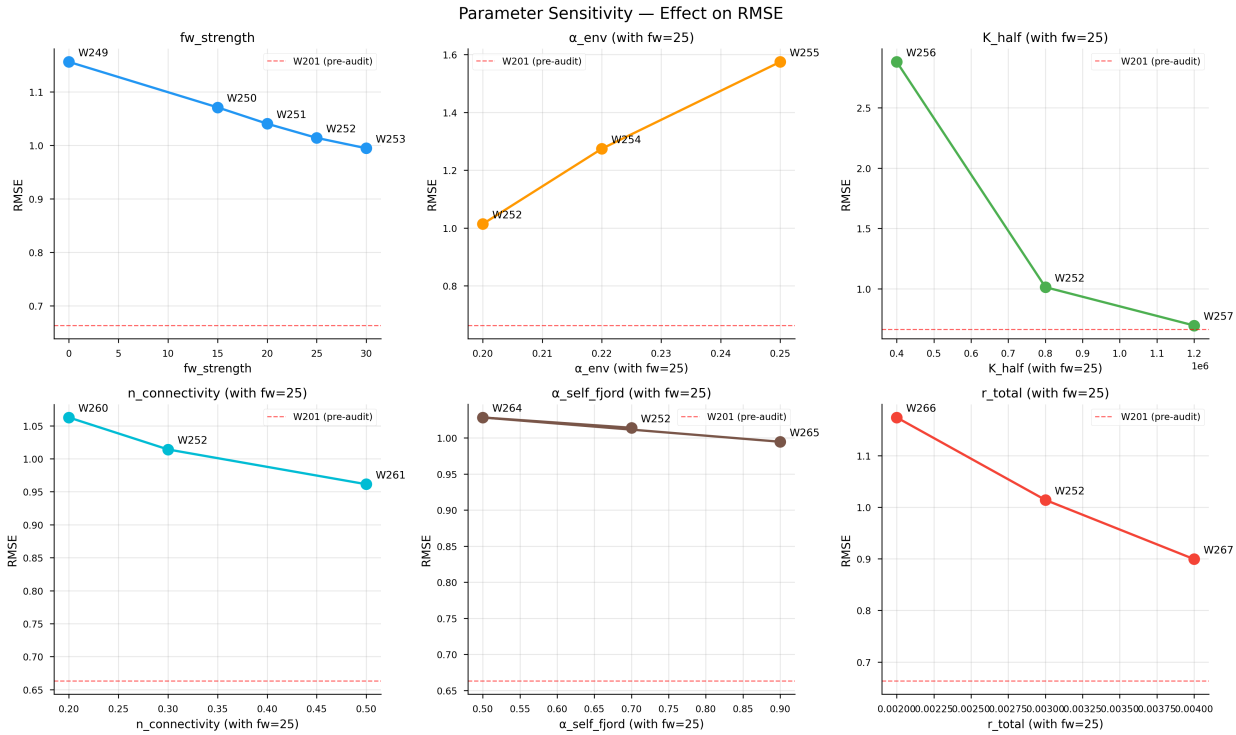

Figure 8: Parameter importance analysis across six swept parameters. Each panel shows the effect of one parameter on RMSE (left) and AK-PWS recovery (right). $K_{\frac{1}{2}}$ is the dominant lever: 400K → 1.2M shifts AK-PWS from <1% to 12.1% and RMSE from 2.88 to 0.70. The f_w sweep (0–30) has a weaker monotonic effect (AK-PWS: 1.8% → 3.2%). $\alpha_{\text{env}} \geq 0.22$ is uniformly suppressive. Network connectivity and r_{total} show intermediate effects.

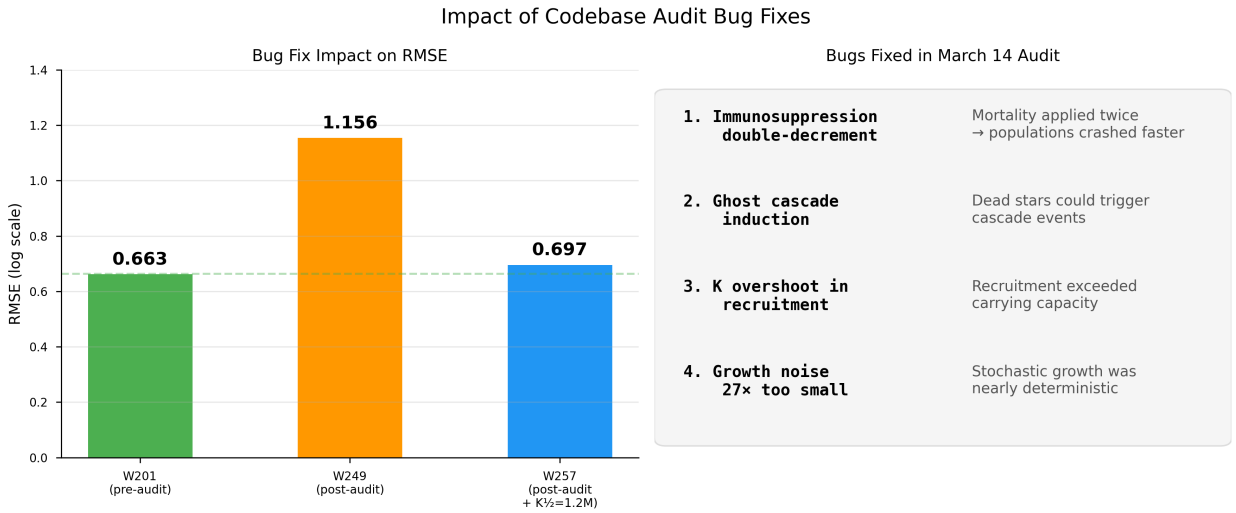

Figure 9: Pre-audit vs. post-audit comparison. The four March 14 bug fixes fundamentally altered model dynamics. The pre-audit W201 baseline achieved RMSE = 0.663, while the equivalent post-audit W249 baseline reaches only RMSE = 1.156 — a 74% degradation. This confirms that prior calibration is not transferable to the corrected codebase. However, W257 (RMSE = 0.697) nearly recovers the pre-audit performance level, demonstrating that $K_{\frac{1}{2}} = 1.2$M compensates for the corrected dynamics.

6 Conclusions & Next Steps

6.1 Key Findings

1. **$K_{\frac{1}{2}}$ is the dominant lever.** Raising $K_{\frac{1}{2}}$ from 400K to 1.2M produces a 6× improvement in AK-PWS recovery (from <1% to 12.1%) and a 4× improvement in RMSE (from 2.88 to 0.70).
2. **Post-audit calibration is viable.** Despite the four bug fixes degrading the baseline RMSE by 74%, W257 recovers to within 5% of the pre-audit best (0.697 vs. 0.663).
3. **The latitude tradeoff persists.** CA-N overshoots its target by 12× in the best configuration. Spatially uniform parameters cannot simultaneously achieve high Alaska recovery and low California recovery.
4. **$\alpha_{\mathbf{env}}$ is suppressive.** Environmental coupling at $\alpha_{\text{env}} \geq 0.22$ degrades RMSE substantially, reducing both northern and southern recovery.

6.2 Recommended Next Steps

1. **Fine-tune** $K_{\frac{1}{2}}$ in the 0.8–1.5M range around the W257 optimum, with 3+ seeds for statistical confidence.
2. **Add spatial modifiers** (latitude-dependent $K_{\frac{1}{2}}$ or region-specific f_w) to suppress southern overshooting while preserving Alaska recovery.
3. **Investigate the $K_{\frac{1}{2}} \times \alpha_{\mathbf{env}}$ interaction**: W259 shows that combining $K_{\frac{1}{2}} = 1.2$M with $\alpha_{\text{env}} = 0.25$ reduces CA-N overshooting (0.64% vs. 1.22%) but also halves AK-PWS recovery (5.2% vs. 12.1%). An intermediate α_{env} value may offer a better tradeoff.
4. **Multi-seed validation**: All results are from a single seed (42). Confirm W257's superiority across 3–5 seeds before committing to the $K_{\frac{1}{2}} = 1.2$M regime.

A Full Metrics

Rank	Config	RMSE	AK-PWS (%)	AK-FN (%)	AK-FS (%)	BC-N (%)	SS-S (%)	JDF (%)	OR (%)	CA-N (%)
	Target	—	*30.0*	*30.0*	*20.0*	*20.0*	*5.0*	*2.0*	*0.25*	*0.10*
1	**W257**	**0.697**	**12.11**	**6.27**	**4.32**	**2.12**	**2.36**	**2.25**	**1.45**	**1.22**
2	W259	0.864	5.17	2.53	1.88	0.97	0.99	0.97	0.74	0.64
3	W267	0.899	4.77	2.31	1.89	0.84	0.95	0.94	0.78	0.76
4	W261	0.961	3.75	1.82	1.45	0.78	0.77	0.77	0.77	0.73
5	W253	0.995	3.20	1.71	1.41	0.54	0.64	0.63	0.54	0.50
6	W265	0.995	3.29	1.76	1.26	0.50	0.68	0.65	0.47	0.42
7	W252	1.014	2.91	1.55	1.27	0.54	0.58	0.58	0.50	0.41
8	W264	1.028	3.20	1.39	1.14	0.51	0.64	0.55	0.49	0.48
9	W251	1.041	2.83	1.34	1.10	0.51	0.61	0.56	0.52	0.46
10	W260	1.063	2.81	1.47	1.07	0.34	0.54	0.49	0.34	0.27
11	W250	1.071	2.49	1.22	0.97	0.47	0.57	0.59	0.53	0.46
12	W249	1.156	1.77	0.82	0.75	0.41	0.56	0.49	0.50	0.49
13	W266	1.174	1.94	1.15	0.71	0.29	0.35	0.36	0.31	0.28
14	W254	1.274	1.83	0.82	0.63	0.20	0.22	0.24	0.21	0.18
15	W263	1.407	1.40	0.55	0.41	0.15	0.15	0.16	0.31	0.24
16	W255	1.575	0.99	0.40	0.30	0.08	0.09	0.08	0.12	0.10
17	W262	1.690	0.89	0.33	0.30	0.04	0.08	0.06	0.05	0.04
18	W268	1.797	0.61	0.21	0.17	0.04	0.06	0.05	0.06	0.06
19	W256	2.879	0.08	0.02	0.02	<0.01	<0.01	<0.01	0.01	<0.01
20	W258	3.548	0.02	0.02	<0.01	<0.01	<0.01	0.00	<0.01	<0.01

Table 3: Full metrics for all 20 configurations, ranked by RMSE. Recovery values are shown as percentages for the 8 calibration regions. Target values are listed in the first data row for reference. W257 (bold) achieves the best overall RMSE and the highest recovery in all northern regions.